Saurabh Yadav

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Education

Koneru Lakshmaiah University **Guntur, Andhra Pradesh, India**
Bachelor of Technology in Computer Science and Engineering | CGPA: 9.61/10.0 *Aug 2023 – May 2027*

- **Specialization:** AI-Driven Language Technology

Technical Skills

Languages: Python, Java, SQL

AI / Machine Learning: Scikit-learn, TensorFlow, Keras, PyTorch, SpaCy, NLP Pipelines, RAG, Prompt Engineering, LLMs, Feature Engineering, Model Fine-Tuning, AST Parsing

Backend: FastAPI, REST APIs, JWT Authentication, Microservices

Databases & Vector Stores: PostgreSQL, MySQL, ChromaDB, FAISS, Redis

Tools & Technologies: Git, GitHub, React.js, OpenRouter, Amazon SageMaker

Projects

ToxiGuard AI – Real-Time Abuse & Toxicity Moderation Platform **Jul 2026**
Tech Stack: *Python, FastAPI, DeBERTa-v3, ONNX Runtime, React 18, Manifest V3, WebSockets, OpenRouter* GitHub | Live Link

- Architected a **3-tier cascaded AI pipeline** (RegEx rules → ONNX DeBERTa → LLM reasoning via OpenRouter), achieving **94.2% ROC-AUC** across 6 toxicity categories while cutting LLM token costs by **65%**.
- Quantized PyTorch Transformer weights into **ONNX Runtime** CPU format, slashing inference latency from 280ms to **<35ms** (**87% speedup**) with $< 10^{-4}$ model output drift.
- Engineered a **Manifest V3 Chrome Extension** using debounced **MutationObservers** for 8 social platforms (X, Reddit, IG, LinkedIn), reducing browser CPU thrashing by **45%** and manual review workload by **70%**.
- Built a production **FastAPI REST/WebSocket API** with JWT auth, SlowAPI rate limiting, and a **React 18** analytics dashboard featuring perturbation-based **XAI token attribution** and batch CSV scanning.

Rigel AI – Intelligent Code Smell Detection Platform **Feb 2026**
Tech Stack: *Python, FastAPI, React.js, AST Parsing, Scikit-learn, NLP, OpenRouter LLM, REST APIs* GitHub | Live Link

- Architected a hybrid **AST-driven static analyzer** and **LLM inference engine** to detect **15+ anti-patterns and performance bottlenecks**, improving code smell detection precision to **94.2%** over standalone LLM baselines.
- Developed asynchronous **FastAPI REST endpoints** with UUID-indexed session persistence for multi-turn AI discussions, reducing multi-file processing latency by **42%** (**<150ms** avg API response time).
- Evaluated ML classifier models on Microsoft's **CodeXGLUE benchmark** and engineered a deterministic **AST syntax validator** (`ast.parse`), reducing invalid AI refactoring suggestions to **<1%**.

Experience

AWS Academy – EduSkills Foundation **Verify**
Artificial Intelligence and Machine Learning Intern (Cohort 12) *Apr 2025 – Jun 2025*

- Designed and deployed **3 supervised learning models** using **AWS SageMaker**:
  - **Sentiment Analysis on E-Commerce Reviews** – Achieved **88%+ accuracy** on **10,000+ product reviews**.
  - **Mine vs. Rock Prediction (Sonar Data)** – Achieved **85%+ accuracy** using **SVM** and **Random Forest** models.
  - **MNIST Digit Classification** – Achieved **97%+ accuracy** using a **3-layer Neural Network**.
- Improved model validation accuracy by **15%** through hyperparameter tuning and cross-validation.
- **Tech Stack:** Python, TensorFlow/Keras, Scikit-learn, Pandas, AWS SageMaker.

Certifications

- Introduction to Artificial Intelligence (IBM, Coursera, Apr 2025) Verify
- AWS Academy Graduate – Machine Learning Foundations (AWS Academy, Jun 2025) Verify
- Supervised Machine Learning: Regression (IBM, Coursera, 2025) Verify

Achievements & Activities

- Solved **500+ DSA problems** across LeetCode, HackerRank, CodeChef, and Codeforces.
- Ranked in the **top 1%** of the Computer Science department with a **CGPA of 9.61/10**.
- Led a **4-member team** at the **Code4Change Hackathon**, building a full-stack ride-booking platform in **24 hours** and securing a place among the **Top 10 finalists**.
- Represented the **Lucknow District U-19 Cricket Team** (2022–2024), demonstrating teamwork and performance under competitive pressure.